INCH-POUND

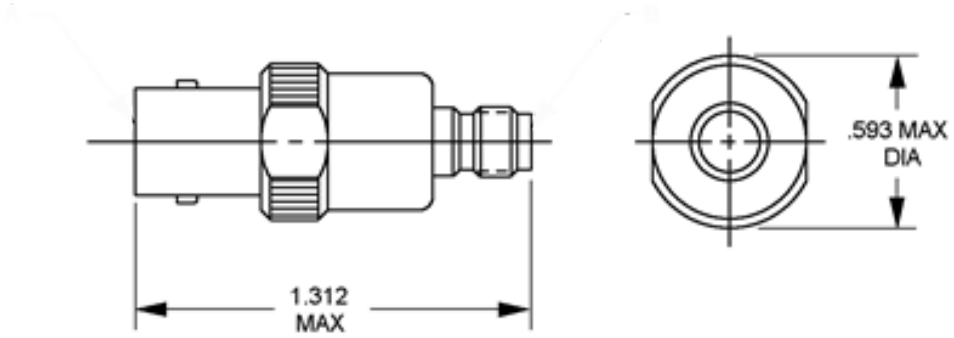
MIL-PRF-55339/46B
w/AMENDMENT 1
03 September 2024
SUPERSEDING
MIL-PRF-55339/46B
28 January 2015

PERFORMANCE SPECIFICATION

ADAPTER, CONNECTOR, COAXIAL, RADIO FREQUENCY, IN-LINE, (BETWEEN SERIES SMA JACK TO SERIES BNC JACK), CLASS 2

This specification is approved for use by all
Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall
consist of this specification sheet and MIL-PRF-55339.



NOTES:
1. Dimensions are in inches.
2. All undimensioned pictorial representations are for reference purposes only.
3. Interface shall be in accordance with MIL-STD-348.

FIGURE 1. General configuration.

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ENGINEERING DATA:

General configuration: See figure 1.

Impedance: 50 ohms, nom.

Working voltage: Sea level – 335 V rms. 70,000 feet – 85 V rms.

Frequency range: 0 to 4 GHz.

Temperature range: -65° to +165°C.

REQUIREMENTS: (installation torque 4 to 6 in. lbs).

Dimensions: See figure 1.

Center contact retention: Axial force – 6 lb, min. Torque – Not applicable.

Force to engage and disengage	Series BNC	Series SMA
Longitudinal force – Torque – (in. lb, max)	3 2.5	Not applicable 2

Mating characteristics:

	Series BNC	Series SMA
Center contact (socket):		
Oversize test pin dia (in min)	.057	.0375 + .0001
Pin finish (microinches)	16	16
Insertion depth (in. min)	.125	030/045
No. of insertions	1	3
Max test pin (insertion force test):		
Steel test pin dia (in. min)	.054	.0370 + .0001
Pin finish (microinches)	16	16
Insertion force (lb, max)	2	3
Insertion depth	--	.050/.075
No. of insertions	1	1
Min test pin (withdrawal force):		
Steel test pin dia (in. max)	.052	.0355 - .0001
Pin finish (microinches)	16	16
Withdrawal force (oz, min)	2	1
Insertion depth	--	.050/.075
No. of withdrawals	1	3

Permeability: <2.0.

Seal:
Pressurized – Not applicable.
Weatherproof – Not applicable.

Insulation resistance: 5,000 megohms, min.

VSWR: 1.30 max at .5 to 4.0 GHz.

RF leakage (total): -55 dB, min, 2 to 3 GHz.

RF insertion loss: .2 dB, max, 3 GHz (.115 $\sqrt{F\ (GHz)}$ dB max tested at 3 GHz).

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Durability: 500 cycles minimum at 12 cycles/min maximum. The connector shall meet the mating characteristics and force to engage and disengage requirements.

Dielectric withstanding: Test voltage – 1,500 V rms, min (sea level).

Contact resistance (milliohms, max):

Contact	Initial	After
Center	4.1 1/	6.0
Outer	2.2	Not applicable

1/ Two center contacts must be mated to the center conductor under test, therefore doubling "center contact" resistance.

Vibration, high frequency: MIL-STD-202-204, test condition D, interruptions – 1 µs, maximum.

Shock (specified pulse): MIL-STD-202-213, test condition I.

Thermal shock: MIL-STD-202-207, test condition C.

Moisture resistance: 200 megohms, min.

Corona level: Voltage – 250 V, min. Altitude – 70,000 feet, min.

RF high potential withstanding voltage: RF voltage – 670 V rms, min. Frequency – 5 MHz, min.

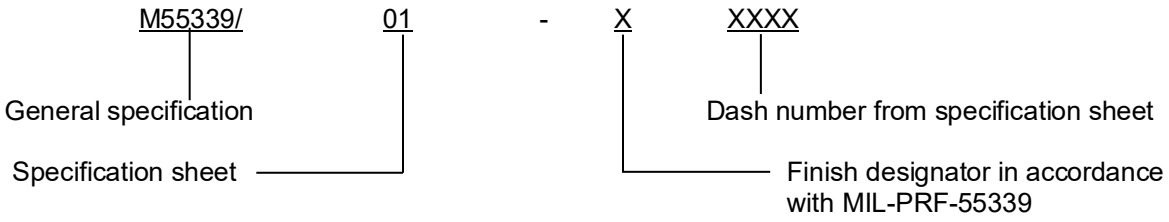
Salt spray (corrosion): Test condition B.

Marking: As specified in MIL-PRF-55339.

Part or Identifying Number (PIN):

M55339/46-30001
M55339/46-50001

PIN configuration:



Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

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Referenced documents: In addition to MIL-PRF-55339, this document references the following:

MIL-STD-202-204
MIL-STD-202-207
MIL-STD-202-213
MIL-STD-348

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA – CC

Preparing activity:
DLA - CC

(Project 5935-2024-198)

Review activities:
Army - AR, AT, EA
Navy - AS, MC, OS
Air Force – 19

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of information above using the ASSIST Online database at https://assist.dla.mil.

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